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BCSE355L - Cloud Architecture Design

Project Title -: Reinforcement Learning-Based Intelligent Queue Optimization Framework for Railway Reservation Systems

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LITERATURE SURVEY

Paper	Method	Dataset	Advantages	Limitations	Research Gap
1) Security-Checking Opening Planning in Transportation Hubs With Reinforcement Learning and Deep Learning Considering Stochastic Passenger Arrival (IEEE 2026)	Hybrid framework using Discrete Event Simulation (DES), Deep Q-Network (DQN), and Multi-Layer Perceptron (MLP) for dynamic queue optimization.	No public dataset is used. The model is trained and evaluated using a simulated dataset generated through Discrete Event Simulation (DES)	Optimizes queues in real time, reducing waiting time and operational cost under uncertain passenger arrivals.	Designed specifically for transportation security checkpoints, not railway reservation systems.	Does not address queue optimization for railway reservation requests, seat allocation, or ticket booking processes.
2) Multi-Agent Deep Reinforcement Learning for Efficient Vehicular Mobility via Federated Fog Computing (ACM)	Multi-Agent Deep Q-Learning (MADQL) integrated with Federated Fog Computing (FFC).	Simulation dataset generated using OpenStreetMap (Gwalior City) and the SUMO simulator.	Reduces vehicle waiting time through decentralized real-time decision making.	Designed for traffic signal control, not railway reservation systems.	Does not optimize railway reservation requests, ticket allocation, or booking queues. Does
3) Responsive Elastic Computing (ACM/IEEE 2022)	SARSA-based Reinforcement Learning with Echo State Networks (ESN) for resource provisioning and job scheduling.	EGEE Grid workload trace containing over 9,000 user jobs collected from the PBS scheduler.	Improves resource utilization and reduces job queuing delay through adaptive RL-based scheduling.	Designed for cloud/grid computing environments, not railway reservation systems.	not address railway reservation queues, ticket booking, or seat allocation optimization.

4) Deep Reinforcement	Transformer-based Deep	Randomly generated EV	Produces faster optimization,	Application is limited to	Does not consider passenger demand
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	Learning-based Multi-Objective Optimization for Mobile Charging Services in Internet of Electric Vehicles (ACM 2023)	Reinforcement Learning (REINFORCE + Rollout Baseline) with weighted multi-objective optimization.	charging requests in a simulated 15×15 km city with 10- and 20-EV scenarios.	better Pareto solutions, strong generalization, and significantly lower computation time than NSGA-II and MOEA/D.	EV charging route optimization and time-window scheduling; no handling of reservation queues, waitlists, or seat allocation.	prediction, ticket booking priority, dynamic waitlist movement, or railway seat allocation under uncertain demand.
5)	Service-Oriented Resource Allocation and Task Scheduling for Wi-Fi and Bluetooth Coexistence in Smart Home IoT Systems (ACM 2025)	Hierarchical Entropy Weight Method (HEWM) for master selection + Proximal Policy Optimization (PPO) for resource allocation + Task Urgency-based Queueing (TUQ) for task scheduling.	Simulation-based Smart Home IoT environment consisting of Bluetooth and Wi-Fi devices, varying network topology, service requests, and communication channel conditions.	Improves system throughput by 3–22%, reduces task delay by 5–36%, dynamically adapts to changing network conditions, efficiently prioritizes urgent tasks, and minimizes communication interference.	Designed specifically for wireless communication resource allocation in IoT systems. It does not model passenger demand, reservation queues, cancellations, berth allocation, or railway booking constraints.	The framework lacks support for real-time railway reservation optimization, waitlist prediction, passenger priority management, cancellation forecasting, and dynamic seat allocation under fluctuating demand.
6)	Intelligent Networking for Energy Harvesting Powered IoT Systems. (ACM 2026)	DeepIoT Routing (Distributed Deep Reinforcement Learning) jointly optimizes energy allocation, multi-hop routing, and topology validation under energy harvesting constraints. It	Simulation-based Energy Harvesting IoT network consisting of multiple EH-powered IoT devices, multi-hop routing topology, varying energy levels, communication	Jointly optimizes routing and resource allocation, improves packet delivery, supports large-scale distributed IoT systems, adapts dynamically to changing energy availability, and outperforms	Jointly optimizes routing and resource allocation, improves packet delivery, supports large-scale distributed IoT systems, adapts	No support for real-time railway reservation systems, dynamic passenger demand prediction, waitlist movement prediction, cancellation forecasting, berth allocation, or AI-driven seat redistribution under fluctuating booking patterns.

	also introduces mathematical topology optimization before DRL-based routing.	n constraints, and dynamic environmental circumstances.	traditional routing techniques through DRL-based decision making.	dynamically to changing energy availability, and outperforms traditional routing techniques through DRL-based decision making.	
7) Interpersonal Communication Interconnection in Media Convergence Metaverse (ACM 2024)	Media Convergence Metaverse Network (MCMN) using a META Controller, Deep Reinforcement Learning (DRL) with LSTM Actor-Critic (DDPG), AI-powered Network Slicing, and Edge-assisted Computing to optimize routing, bandwidth, traffic engineering, and QoS dynamically.	Simulation-based Metaverse Network Environment consisting of heterogeneous network topologies, edge servers, traffic demand matrices, telemetry data, QoS parameters (latency, packet loss), and dynamic routing scenarios. No real-world public dataset is used.	Adaptive AI-based optimization capable of learning dynamic traffic patterns, minimizing link utilization, improving QoS, enabling intelligent network slicing, scalable decentralized control, and continuous learning through DRL and LSTM-based decision making.	Domain-specific to Metaverse networking. It focuses only on communication routing and QoS management and does not address reservation management, passenger demand prediction, seat allocation, cancellations, waitlist optimization, or railway operational constraints.	No mechanism for intelligent railway reservation optimization, dynamic berth allocation, waitlist movement prediction, cancellation forecasting, passenger prioritization, or real time seat redistribution based on booking behavior and train occupancy.

8)	Resource Management for Mission-Critical Applications in Edge Computing: Systematic Review on Recent Research and Open Issues (ACM 2025)	Comprehensive Resource Management Framework covering Resource Allocation, Task Offloading, Application Placement, QoS-aware Scheduling, Deep Reinforcement Learning (DRL), Deep Learning (DL), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Federated Learning (FL), Heuristic Algorithms, Game Theory, Dynamic Voltage and Frequency Scaling (DVFS), and Mathematical Optimization for mission-critical edge environments.	Systematic Literature Review Dataset consisting of 154 research papers (2018–2023) collected from IEEE, ACM, Springer, Elsevier, Wiley, MDPI, and Google Scholar. It is a survey paper and does not use a single benchmark dataset but synthesizes findings across multiple experimental environments.	Provides a comprehensive taxonomy of resource management objectives (latency, energy, reliability, availability, mobility, fault tolerance, cost, multi-objective optimization), compares numerous AI techniques, identifies QoS trade-offs, and highlights open research challenges for future intelligent systems. The taxonomy (Figure 4) clearly categorizes resource management strategies and objectives.	Being a survey paper, it does not propose or validate a new algorithm. The work focuses on edge computing applications such as healthcare, IoT, UAVs, smart grids, and vehicular systems, without addressing reservation systems, passenger demand prediction, berth allocation, cancellations, or railway-specific constraints	No AI framework exists for intelligent railway reservation optimization that jointly considers booking demand, cancellations, waitlist prediction, berth allocation, dynamic seat redistribution, and passenger prioritization using multi-objective optimization under real-time constraints
9)	Dynamic Sizing of Cloud-Native Telco Data Centers With Digital Twin and Reinforcement Learning (IEEE 2024)	Digital Twin-assisted Reinforcement Learning (PPO) integrated with Integer Linear Programming (ILP) and the Rank-Drain-Observe (RDO)	Real-life Kubernetes-based telco edge data center testbeds combined with a Digital Twin profiling dataset generated	Reduces operational costs by dynamically optimizing active resources using RL while enabling safe and efficient training through a Digital Twin.	Designed specifically for cloud-native data center resource management and does not address railway	Does not consider passenger demand, ticket booking, waitlist management, cancellation prediction, or railway reservation optimization.

	algorithm for dynamic data center sizing.	using the Kwok emulator.		reservation queues or seat allocation.	
10) AI: It's All About Inference (ACM Queue 2025)	Inference optimization using pruning, knowledge distillation, quantization, KV-cache optimization, speculative decoding, multi-token prediction (MTP), LayerSkip, continuous batching, distributed inference, Mixture of Experts (MoE), and test-time compute scaling.	No dedicated benchmark dataset. The article surveys techniques evaluated on LLMs (e.g., Llama-2, Llama-3, DeepSeek, GPT-based models) and discusses inference frameworks such as vLLM and SGLang.	Provides a comprehensive overview of state-of-the-art inference optimization methods that reduce latency, memory usage, and deployment cost while improving scalability and real-time performance.	Survey article rather than a domain-specific implementation. It focuses on LLM inference and does not address application-specific optimization such as ticket reservation, passenger flow prediction, or railway seat allocation.	Does not explore how inference optimization can be integrated with transportation systems, reservation platforms, or reinforcement learning for dynamic resource allocation.

11) Oikonomos-II+: A Reinforcement-Learning, Cloud-Resource Recommender for HPC & AI Workloads (IEEE Transactions and Cloud Computing 2026)	Hybrid Reinforcement Learning framework (Oikonomos-II+) combining Neural-LinUCB contextual multi-armed bandits, Gaussian Process Regression (GPR), Variational Autoencoder (VAE), RHO-LOSS data selection, and inverse-variance weighting for efficient online cloud-resource recommendation without historical data.	Evaluation on six benchmark applications (3 HPC + 3 AI workloads), including SimHH, WhiskEras 2.0, Oceananigans, MNIST MLP, CIFAR-10 CNN, and Sunspot RNN, executed across multiple Amazon EC2 instance types.	Learns online without requiring historical training data or repeated trial executions, balances exploration and exploitation, adapts to changing workloads, reduces cloud cost and execution time, and converges to optimal resource recommendations.	Designed specifically for cloud HPC/AI resource allocation. It does not consider domain-specific constraints such as passenger demand prediction, reservation priorities, cancellation behavior, waitlist management, or fairness in transportation systems.	The framework is not adapted for real-time transportation decision-making or reservation optimization. It also lacks integration with passenger behavior analytics and dynamic seat allocation policies.
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12) Open RAN Slicing for MVNOs with Deep Reinforcement Learning (IEEE 2023)	Two-Level RAN Slicing (2LRS) framework combining Game-Theoretic Matching (MSDA) for long-term resource allocation and Double Deep Q-Network (DDQN)-based Deep Reinforcement Learning for short-term user-level resource allocation. The optimization is modeled as two NP-hard problems solved through matching algorithms and DRL.	Simulation-based evaluation using a 5G Open RAN environment consisting of an MNO, multiple MVNOs, eMBB and URLLC users, radio resource blocks (RBs), varying channel conditions, QoS constraints, and O-RAN architecture. No public benchmark dataset is used; experiments are performed through network simulations.	Efficiently handles dynamic resource allocation using hierarchical optimization, considers individual user QoS requirements, combines long-term and short-term decision making, reduces resource allocation delay, and outperforms existing centralized approaches in simulation.	Efficiently handles dynamic resource allocation using hierarchical optimization, considers individual user QoS requirements, combines long-term and short-term decision making, reduces resource allocation delay, and outperforms existing centralized approaches in simulation.	The proposed DRL framework is not adapted for transportation reservation systems where user preferences, booking history, cancellations, seasonal demand, and fairness policies significantly affect allocation decisions.
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13) Urban Computing in the Era of Large Language Models (ACM 2025)	Comprehensive Survey of LLM-based Urban Computing Frameworks. The paper categorizes LLM applications into five functional roles—Encoder, Predictor, Enhancer, Assistant, and Agent—across seven domains including Intelligent Transportation Systems, Urban Mobility, Travel Planning, Public Safety, Environmental Monitoring, Urban Planning, and Resource Optimization. It summarizes techniques such as Prompt Engineering, Chain-of-Thought (CoT), RLHF, PEFT (LoRA), Instruction Tuning, and Retrieval-Augmented Generation (RAG).	No single dataset. Being a survey, it reviews numerous benchmark datasets and LLM applications across multiple urban domains including traffic prediction, travel planning, mobility prediction, energy forecasting, geospatial intelligence, and transportation systems instead of proposing a new dataset.	Provides a comprehensive taxonomy of LLM applications, identifies state-of-the-art models across transportation and mobility, summarizes datasets and evaluation methods, discusses implementation patterns, and highlights future research directions. It serves as an excellent foundation for designing intelligent transportation systems.	This is a survey paper rather than a proposed solution. It does not implement or evaluate an actual railway reservation model, lacks experimental validation for reservation systems, and does not address railway-specific challenges such as seat allocation, waitlist optimization, cancellation, berth preference, quota management, or fare optimization.	Current LLM research in transportation focuses primarily on traffic prediction, urban mobility, and travel planning, leaving intelligent railway reservation systems with dynamic seat allocation, passenger preference modeling, waitlist prediction, and reservation optimization largely unexplored.
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14) Deep Reinforcement Learning-Based AI Task Offloading in Resource-Constrained IIoT Computing Environments	LHC-DQN (LSTM-Enhanced Hierarchical Classification with Deep Q-Network) using a two-layer cloud-edge-device architecture with three LSTM-DQN agents for intelligent task offloading.	Simulation-based IIoT cloud-edge-device environment consisting of cloud servers, enterprise edge servers, distributed IoT devices, and heterogeneous AI tasks (training and inference).	Improves task completion rate, reduces execution latency, and enhances resource utilization through hierarchical LSTM-based reinforcement learning.	Designed specifically for Industrial IoT task offloading and does not address railway reservation, passenger demand, seat allocation, cancellations, or waitlist management.	Does not optimize railway reservation queues, ticket booking priorities, dynamic berth allocation, RAC/waitlist movement, or passenger demand prediction using reinforcement learning.
15) Improving Storage Systems Using Machine Learning (ACM 2023)	KML (Kernel Machine Learning) framework using Neural Networks and Decision Trees to dynamically optimize storage parameters such as readahead and NFS read-size (rsize).	Trace-based storage workload dataset collected from RocksDB benchmarks, NVMe-SSD, SATA-SSD, and NFS environments using LTTng kernel tracing (over 20 GB of traces).	Dynamically adapts to changing workloads, achieving up to 2.3× improvement in readahead throughput and 15× improvement in NFS throughput with very low CPU (<0.2%) and memory (~4 KB) overhead.	Limited to storage-system parameter tuning and does not address reservation queues, passenger demand, ticket allocation, waitlist management, or reinforcement learning-based optimization.	Does not explore intelligent queue optimization, dynamic resource allocation for reservation systems, or sequential decision making using Reinforcement Learning in transportation applications.

Paper	Problem Improvement
1) Security-Checking Opening Planning in Transportation Hubs With Reinforcement Learning and Deep Learning Considering Stochastic Passenger Arrival	Adapt the RL-based framework to railway reservation systems by optimizing reservation queues and seat allocation in real time.
2) Multi-Agent Deep Reinforcement Learning for Efficient Vehicular Mobility via Federated Fog Computing	Extend the multi-agent RL framework to railway reservation systems for dynamic ticket allocation and reservation queue optimization.
3) Responsive Elastic Computing	Adapt the RL-based scheduling framework to railway reservation systems for intelligent reservation queue and seat allocation management.
4) Deep Reinforcement Learning-based Multi-Objective Optimization for Mobile Charging Services in Internet of Electric Vehicles	Extend the DRL framework to railway reservation by incorporating passenger demand forecasting, dynamic waitlist prioritization, berth allocation, cancellation prediction, and real-time seat optimization using historical reservation data.
5) Service-Oriented Resource Allocation and Task Scheduling for Wi-Fi and Bluetooth Coexistence in Smart Home IoT Systems	Adapt the PPO-based scheduling framework for Indian Railway reservation by integrating passenger demand forecasting, cancellation prediction, RAC/Waitlist prioritization, berth allocation optimization, and real-time seat redistribution using historical booking data and live reservation updates.
6) Intelligent Networking for Energy Harvesting Powered IoT Systems	Extend the distributed DRL framework to Indian Railway Reservation by integrating passenger demand forecasting, cancellation prediction, berth allocation optimization, RAC/Waitlist prioritization, and dynamic seat redistribution using historical reservation data and real-time booking updates.
7) Interpersonal Communication Interconnection in Media Convergence Metaverse	Extend the DRL + LSTM optimization framework to Indian Railway Reservation by replacing network traffic states with passenger booking states, incorporating demand forecasting, cancellation prediction, berth allocation optimization, RAC/Waitlist prioritization, and real-time seat redistribution using historical reservation data and live booking information.
8) Resource Management for Mission-Critical Applications in Edge Computing: Systematic Review on Recent Research and Open Issues	Develop a hybrid AI framework combining Demand Forecasting (LSTM/Transformer) + Deep Reinforcement Learning + Multi-objective Optimization to dynamically allocate railway seats, predict cancellations, optimize RAC/Waitlist movement, improve berth utilization, minimize passenger waiting time, and maximize reservation efficiency using historical IRCTC booking data and live reservation updates.
9) Dynamic Sizing of Cloud-Native Telco Data Centers With Digital Twin and Reinforcement Learning	Adapt the Digital Twin + RL framework to railway reservation systems for intelligent queue management, dynamic seat allocation, and real-time booking optimization.
10) AI: It's All About Inference	Apply efficient inference techniques (quantization, speculative decoding, continuous batching, MoE, and test-time scaling) to AI-based railway reservation systems.

	systems for faster booking responses, intelligent seat allocation, and reduced server latency during peak booking periods.
11) Oikonomos-II+: A Reinforcement-Learning, Cloud-Resource Recommender for HPC & AI Workloads	Extend the contextual RL framework to railway reservation by treating seat allocation as a contextual bandit problem. Incorporate passenger demand forecasting, waitlist prediction, cancellation probabilities, and real-time booking behavior to recommend optimal seat allocation while minimizing waiting time and maximizing occupancy.
12) Open RAN Slicing for MVNOs with Deep Reinforcement Learning	Extend the hierarchical DRL framework to railway reservation by treating seat allocation as a sequential decision-making problem. Integrate passenger demand forecasting, cancellation prediction, waitlist prioritization, coach occupancy balancing, and dynamic reservation policies to maximize seat utilization while reducing waiting-list length and improving passenger satisfaction.
13) Urban Computing in the Era of Large Language Models	Develop an LLM-powered Smart Railway Reservation System by combining LLM reasoning with demand forecasting, passenger preference analysis, reinforcement learning for dynamic seat allocation, RAG for railway policy retrieval, and predictive models for cancellations and waitlist confirmation. This would optimize seat utilization, reduce waitlist length, improve passenger satisfaction, and support real-time reservation decision-making.
14) Deep Reinforcement Learning-Based AI Task Offloading in Resource-Constrained IIoT Computing Environments	Extend the LHC-DQN framework to railway reservation systems by integrating passenger demand forecasting, cancellation prediction, dynamic seat allocation, RAC/waitlist prioritization, and real-time reservation queue optimization.
15) Improving Storage Systems Using Machine Learning	Extend the adaptive ML framework by incorporating Deep Reinforcement Learning to optimize railway reservation queues, seat allocation, cancellation prediction, and waitlist movement using historical IRCTC booking data and real-time reservation requests.